



STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher: Dr. Aymen TRIGUI

Course Code : 462 CCS-3

Student ID: 445801958

Section Number : 904/907

S.No

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LAB ACTIVITY - 2

(1.5 Marks)

The file foodtruck.xlsx contains the dataset for our simple linear regression problem. The first column is the population of a city and the second column is the profit of a food truck in that city. A negative value for profit indicates a loss.

In this part of the Lab, you will implement linear regression with one variable to predict profits for a food truck.

Requirements:

1. Data visualization
2. Compute the parameters of the optimal Linear regression model
 w_{opt} , b_{opt} , SS_{res} , SS_{total} and R^2
3. Evaluate the Model using LR_Eval
4. Subplot the data and the computed LR model

Dataset shape: (97, 2)

Model parameters:

- w_{opt} : 1.193034
- b_{opt} : -3.895781
- SS_{res} : 868.532447
- SS_{total} : 2914.847052
- R^2 : 0.702032

Sample predictions:

Population	Actual	Predicted
6.1101	17.5920	3.393774
5.5277	9.1302	2.698951
8.5186	13.6620	6.267196
7.0032	11.8540	4.459272
5.8598	6.8233	3.095158



Marks Obtained	/1.5
Teacher's Signature:	